COHOMOLOGY ON THE SET OF SPECIALISATIONS OF A QUASI-COMPACT SUBSET AS A COLIMIT OVER ITS NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The second paragraph is the author's own 'The same proof works for part (2) except we use ...'; the sentence is reproduced verbatim and the variant result is carried as a navigation stub.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-cohomology-of-neighbourhoods-of-closed is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Structure of the set of specialisations of a quasi-compact subset as an inverse limit of quasi-compact open neighbourhoods. Topology, Lemma topology-lemma-make-spectral-space, invoked at proof lines 3884-3885. Supplies the presentation of W as an inverse limit.
- Subspaces appearing as quasi-compact opens are spectral. Topology, Lemma topology-lemma-spectral-sub, invoked at proof lines 3886-3887. Supplies that the colimit formula is applicable to each neighbourhood.
- Colimit formula for cohomology on a spectral space presented as an inverse limit. Same chapter, Lemma lemma-colimit, invoked at proof lines 3888. Supplies both statements of the lemma.
- The same structure theorem for the complement of a constructible subset. Topology, Lemma topology-lemma-make-spectral-space-minus, invoked at proof lines 3889-3890. Supplies the second statement.

Source reference. Original lemma 0A37: sources/stacks-cohomology/pinned-original-source.tex, source lines 3360–3360. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0A31: sources/stacks-chow/tags.txt, source lines 12724–12724. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 0AP0: sources/stacks-chow/tags.txt, source lines 13458–13458. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0902: sources/stacks-chow/tags.txt, source lines 11395–11395. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a spectral space. Let $\mathcal{F}$ be an abelian sheaf on X. Let $E \subset X$ be a quasi-compact subset. Let $W \subset X$ be the set of points of X which specialize to a point of E.*

(1) $H^p(W, \mathcal{F}|_W) = \operatorname{colim} H^p(U, \mathcal{F})$ *where the colimit is over quasi-compact open neighbourhoods of E,*
(2) $H^p(W \setminus E, \mathcal{F}|_{W \setminus E}) = \operatorname{colim} H^p(U \setminus E, \mathcal{F}|_{U \setminus E})$ *if E is a constructible subset.*

Proof. From Topology, Lemma 0A31 we see that $W = \lim U$ where the limit is over the quasi-compact opens containing E. Each U is a spectral space by Topology, Lemma 0902. Thus we may apply Lemma 0A37 to conclude that (1) holds. The same proof works for part (2) except we use Topology, Lemma 0AP0. □